

Assignment 7: GLMs (Linear Regressios, ANOVA, & t-tests)

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A07_GLMs.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

!! AI generated code should NOT be used in this assignment

Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (NTL-LTER_Lake_ChemistryPhysics_Raw.csv). Set date columns to date objects.
2. Build a ggplot theme and set it as your default theme.

#1

```
library(here); library(tidyverse); library(agricolae); library(ggplot2); library(lubridate)
```

```
## here() starts at /home/guest/R/EDE_Fall2025/EDE_Fall2025
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.5
```

```
## v forcats    1.0.0      v stringr    1.5.1
```

```
## v ggplot2    4.0.0      v tibble     3.2.1
```

```
## v lubridate  1.9.3      v tidyr      1.3.1
```

```
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```

here()

## [1] "/home/guest/R/EDE_Fall2025/EDE_Fall2025"

NTL_LTER_raw = read.csv(
  file = here('Data', 'Raw', 'NTL-LTER_Lake_ChemistryPhysics_Raw.csv'),
  stringsAsFactors = TRUE
)
NTL_LTER_raw$sampleddate = as.Date(NTL_LTER_raw$sampleddate, format = "%m/%d/%y")

#2
mytheme = theme_bw()+
  theme(plot.title = element_text(hjust = 0.5),
        plot.subtitle = element_text(hjust = 0.5),
        legend.position = "bottom"
  )
set_theme(mytheme)

```

Simple regression

Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

3. State the null and alternative hypotheses for this question:

Answer:

H_0 : Mean July temperature is the same at all depths across lakes.

H_a : Mean July temperature differs with depth and/or across lakes.

4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:
 - Only dates in July.
 - Only the columns: `lakename`, `year4`, `daynum`, `depth`, `temperature_C`
 - Only complete cases (i.e., remove NAs)
5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

```

#4
NTL_LTER = NTL_LTER_raw %>%
  filter(month(sampleddate) == 7) %>%
  select('lakename', 'year4', 'daynum', 'depth', 'temperature_C') %>%
  drop_na()

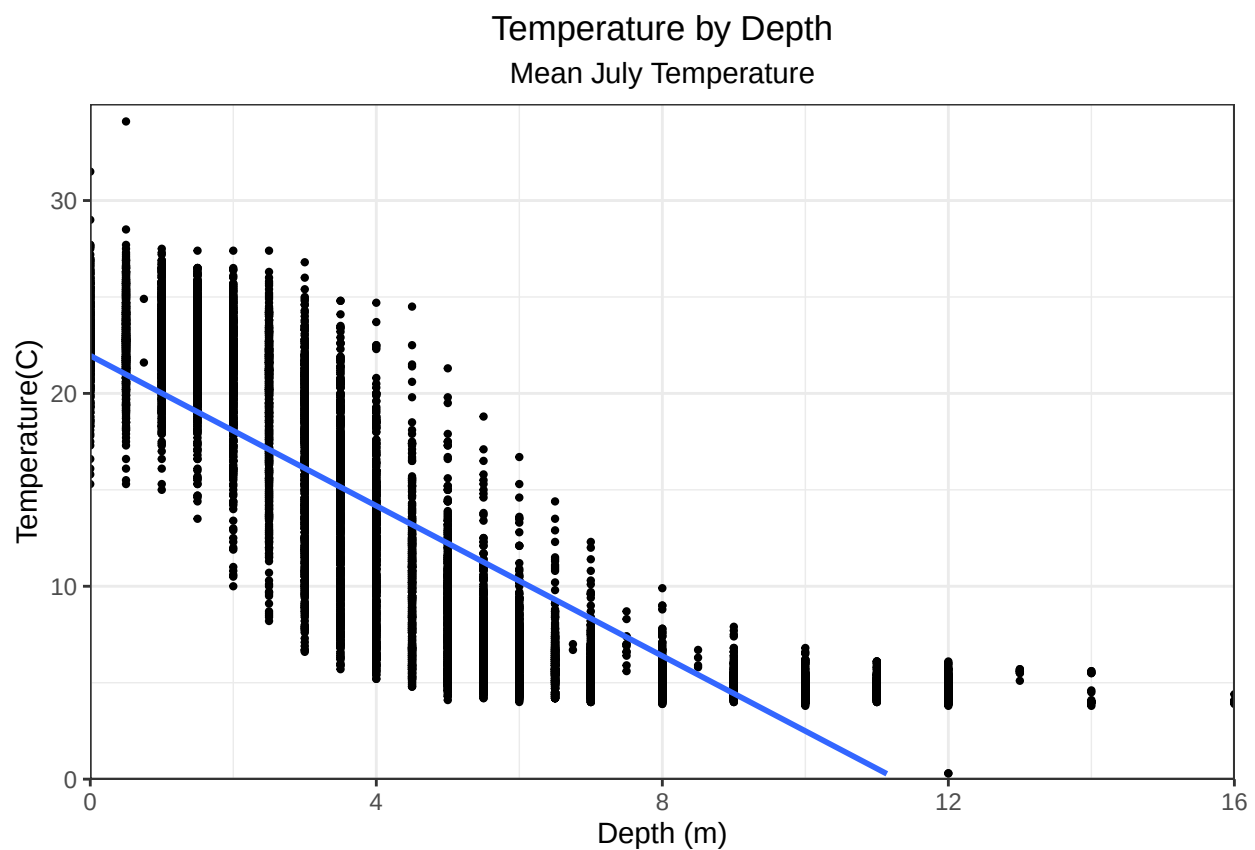
#5
ggplot(NTL_LTER, aes(x = depth, y = temperature_C)) +
  geom_point(size = 0.8) +
  geom_smooth(method = "lm", se = FALSE) +
  scale_y_continuous(expand = c(0, 0), limits = c(0, 35)) +

```

```
scale_x_continuous(expand = c(0,0))+
labs(
  title = "Temperature by Depth",
  subtitle = "Mean July Temperature",
  x = "Depth (m)",
  y = "Temperature(C)",
)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 24 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest about anything about the linearity of this trend?

Answer: The blue regression line suggests a strong negative trend overall. The distribution of points shows some curvature or nonlinearity — temperatures decrease rapidly in the upper few meters, then level off at greater depths.

7. Perform a linear regression to test the relationship and display the results.

```

#7
LR = lm(data = NTL_LTER, temperature_C ~ depth)
summary(LR)

##
## Call:
## lm(formula = temperature_C ~ depth, data = NTL_LTER)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.5173 -3.0192  0.0633  2.9365 13.5834
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 21.95597    0.06792   323.3  <2e-16 ***
## depth       -1.94621    0.01174  -165.8  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.835 on 9726 degrees of freedom
## Multiple R-squared:  0.7387, Adjusted R-squared:  0.7387
## F-statistic: 2.75e+04 on 1 and 9726 DF, p-value: < 2.2e-16

```

8. Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

Answer: Depth explains about 73.9% of the variability ($R^2 = 0.7387$). The result is based on 9,726 degrees of freedom, showing a large sample size and reliable estimates. The relationship is highly significant because ($p < 2.2 \times 10^{-16}$). For every 1-meter increase in depth, the mean July temperature decreases by approximately 1.95°C.

Multiple regression

Let's tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

9. Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.
10. Run a multiple regression on the recommended set of variables.

```

#9
GLM = lm(data = NTL_LTER,
          temperature_C ~ year4 + daynum + depth)
step(GLM)

```

```
## Start: AIC=26065.53
## temperature_C ~ year4 + daynum + depth
##
##           Df Sum of Sq    RSS   AIC
## <none>                 141687 26066
## - year4    1         101 141788 26070
## - daynum   1        1237 142924 26148
## - depth    1       404475 546161 39189

##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = NTL_LTER)
##
## Coefficients:
## (Intercept)      year4      daynum      depth
##    -8.57556     0.01134     0.03978    -1.94644
```

```
#10
summary(GLM)
```

```
##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = NTL_LTER)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.6536 -3.0000  0.0902  2.9658 13.6123
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -8.575564   8.630715  -0.994  0.32044
## year4         0.011345   0.004299   2.639  0.00833 **
## daynum        0.039780   0.004317   9.215 < 2e-16 ***
## depth        -1.946437   0.011683 -166.611 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.817 on 9724 degrees of freedom
## Multiple R-squared:  0.7412, Adjusted R-squared:  0.7411
## F-statistic: 9283 on 3 and 9724 DF, p-value: < 2.2e-16
```

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

Answer:

Analysis of Variance

12. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality)

or similar variances.) Create two sets of models: one expressed as an ANOVA models and another expressed as a linear model (as done in our lessons).

#12

13. Is there a significant difference in mean temperature among the lakes? Report your findings.

Answer:

14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a `geom_smooth` (`method = "lm"`, `se = FALSE`) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

#14.

15. Use the Tukey's HSD test to determine which lakes have different means.

#15

16. From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

Answer:

17. If we were just looking at Peter Lake and Paul Lake. What's another test we might explore to see whether they have distinct mean temperatures?

Answer:

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does the test say? Are the mean temperatures for the lakes equal? Does that match your answer for part 16?

Answer: